

Yash Rawal

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Professional Summary

Applied AI & Machine Learning Engineer serving a **1M+ user scale** by architecting enterprise agentic platforms. Specialized in PyTorch foundation model pretraining (~85M parameters), custom Triton/CUDA attention kernels, and self-evolving agent loops. Active contributor to Spring AI, bridging core machine learning research with high-performance production systems.

Skills

- Languages & Backend:** Python, Java, SQL, FastAPI, Spring Boot, PostgreSQL, REST APIs
- AI & ML Engineering:** Model Training & SFT, Agentic AI, Distributed Training & Inference, KV Caching, MCP, Vector Embeddings
- AI & ML Frameworks:** PyTorch, Triton, CUDA, HIP, vLLM, Hugging Face, Spring AI, LangGraph, CrewAI
- Platforms & Models:** Claude, Amazon Bedrock, Google Vertex AI, Azure OpenAI, Qwen, TinyStories, Meta Llama & Open Weights
- Infrastructure & Security:** Docker, Kubernetes, Linux, CI/CD, LLM-as-Judge, Red Teaming, PII Sanitization, WebSockets

Work Experience

TalenTeam Ltd., UK (via Bigscal Technologies)
AI/ML Engineer Oct 2024 - Present

- Remote
- Deployed a production-scale agentic platform and multi-agent systems serving **1M+ active ERP users** across 120+ enterprise tenants.
 - Designed a capability contract layer to execute nested profiles inside a single ReAct loop, reducing execution overhead by **3x** and token consumption by **70%**.
 - Trained domain-specific **Qwen2.5-7B** models on A100 GPUs and fine-tuned E5 embeddings, writing custom CUDA and Triton kernels to optimize low-latency tensor operators.
 - Developed the Prompt Lab evaluation control plane to collect reasoning trajectories and human preference annotations, establishing a scalable **Human Feedback (RLHF) Pipeline**.
 - Implemented a hierarchical RAG pipeline indexing 10,000+ technical documents using GROBID, pdfplumber, and PostgreSQL/PGVector, achieving a 90%+ Recall@10 retrieval rate.
 - Secured agentic workflows against OWASP LLM vulnerabilities using phantom-token context propagation, runtime tool monitoring, and trained token-classification models for automated PII redaction.

Healthray (Bigscal Technologies)
AI/ML Engineer Intern Apr 2024 - Sep 2024

- Surat, India
- Engineered clinical summarization pipelines for 116K+ medical records with 4-bit quantized Llama 3, and designed real-time WebSocket audio streaming for Indic ASR (Whisper, Vosk, NeMo).
 - Researched audio digital signal processing (STFT, Mel-spectrograms, MFCC feature extraction via librosa/torchaudio) and CNN audio architectures (VGGish, ResNet), compiling custom Indic LJSpeech-format datasets.

Projects

Mini Language Model (~85M Parameter Architecture) | Personal Research

Pretrained an ~85M-parameter transformer to ~1.43 validation loss on TinyStories; engineered FragmentStream chunked attention to eliminate $O(T^2)$ VRAM overhead on commodity GPUs and custom Triton kernels with KV caching for low-latency inference.

Python, PyTorch, CUDA, Triton, Tesla P100/T4

Modular Agent Harness & Memory System ("Brain") | Open Source

Principal Creator & Architect of an open-source agentic framework designed to automatically generate and evaluate high-performance CUDA/HIP workloads, implementing a formal ReAct loop (Thought-Action-Observation) with tactical tool routing.

Java, Spring Boot, PostgreSQL, WebSockets

Spring AI (spring-projects/spring-ai) | Open Source

Merged PR #5711 to propagate reasoningContent from AWS Bedrock and OpenAI-compatible models, and resolved Converse API prompt-caching TTL latency consistency.

Java, Spring Boot

Education

Master of Computer Applications (Artificial Intelligence & Machine Learning) - Parul University 2022 - 2024

Certifications

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| Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft |
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